Pikachu Metrics 数据分析 metrics_walk_back_20260820_194353_with_target

FILE
metrics_walk_back_20260820_194353_with_target.csv

NETWORK
TransitionDown

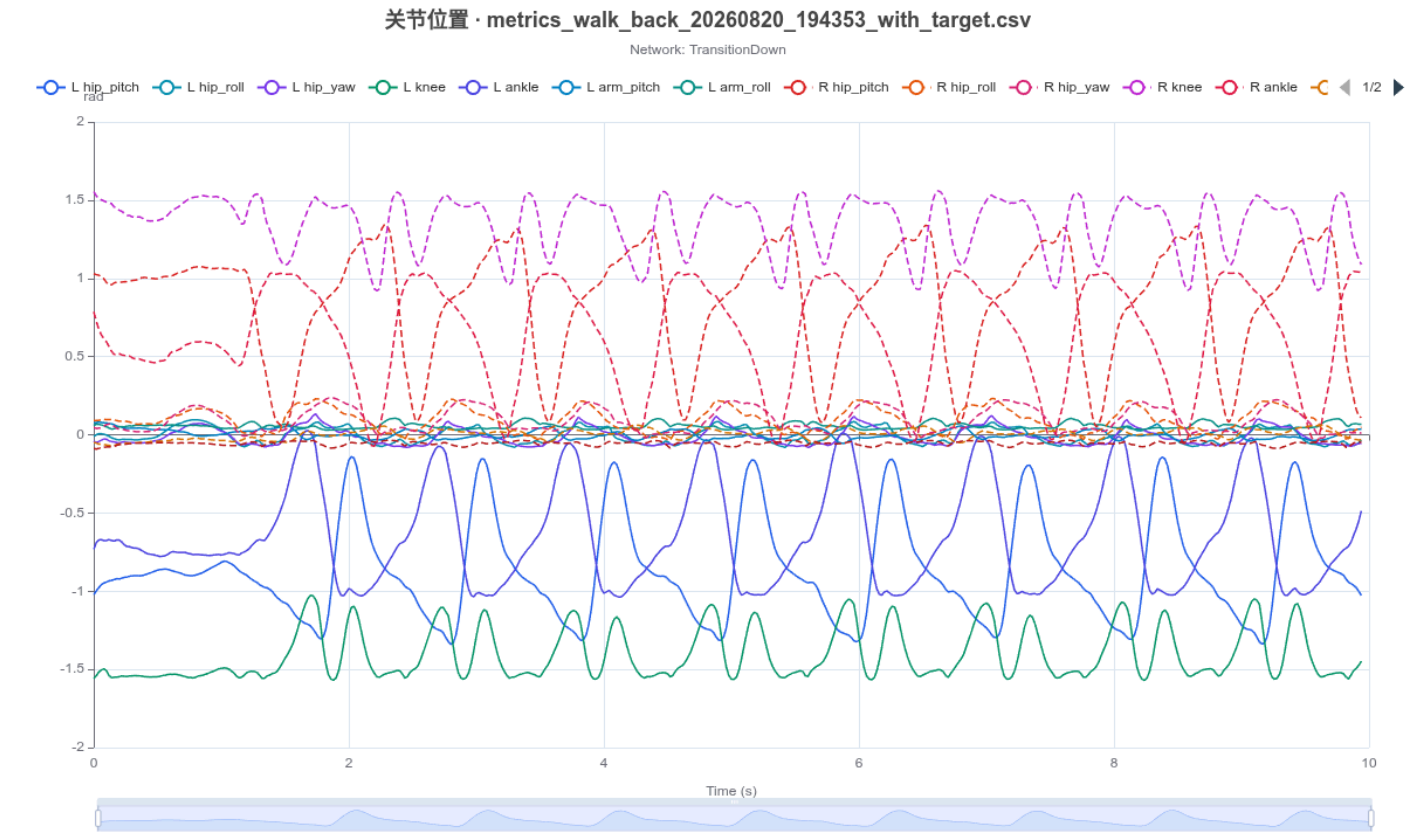
SAMPLES
498

DURATION
9.940 s

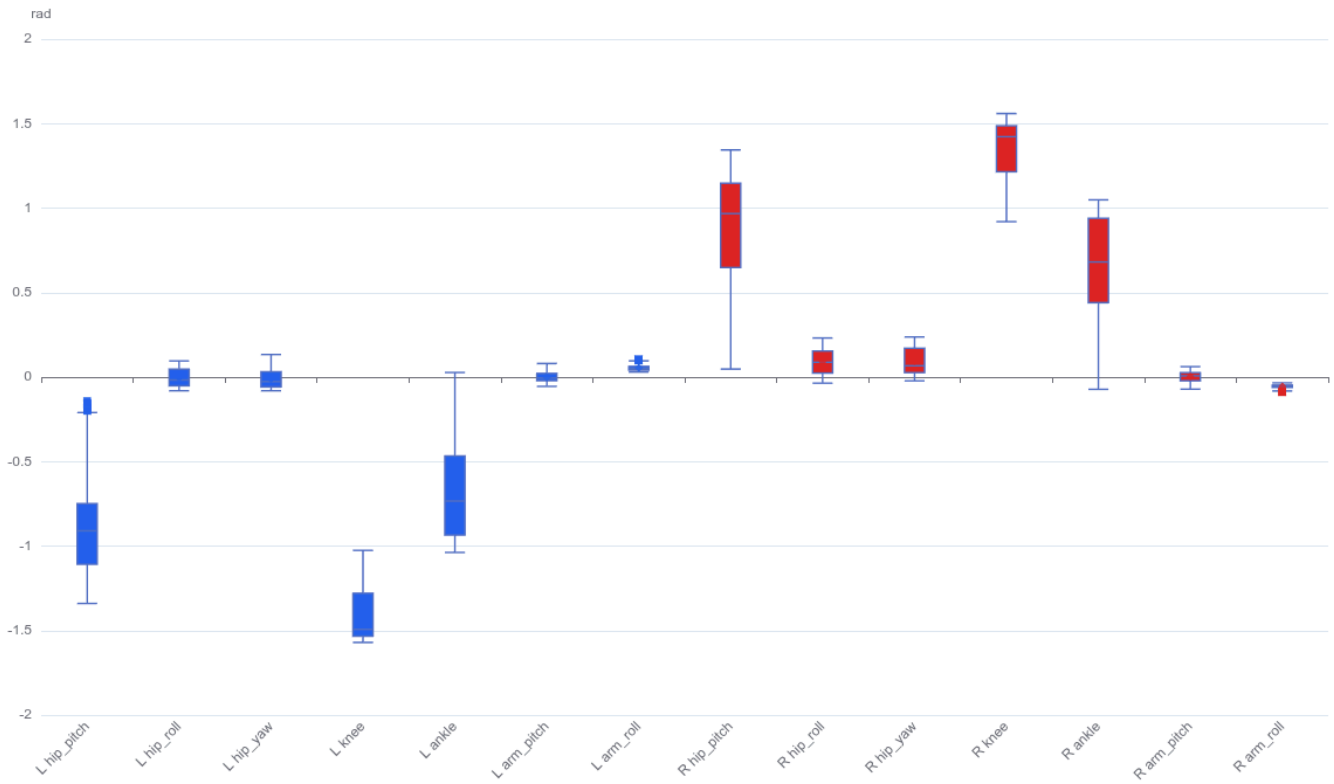
ENVIRONMENTS
1

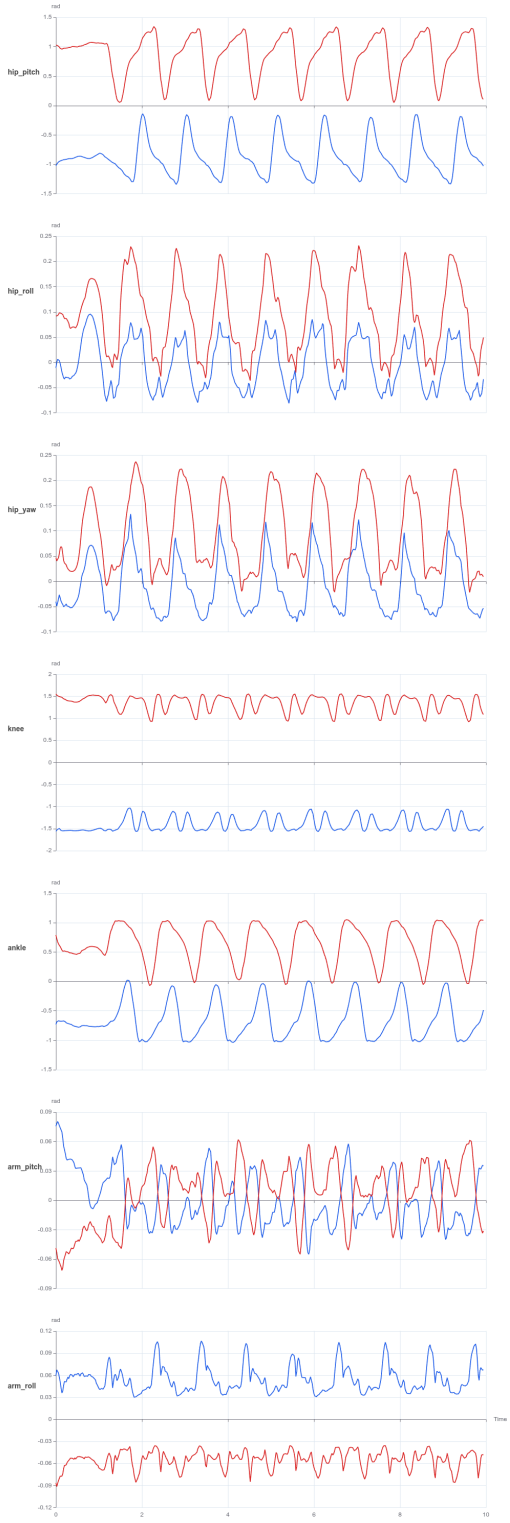
时间区间: 0.000 — 9.940 s · 仅含已勾选字段

关节位置

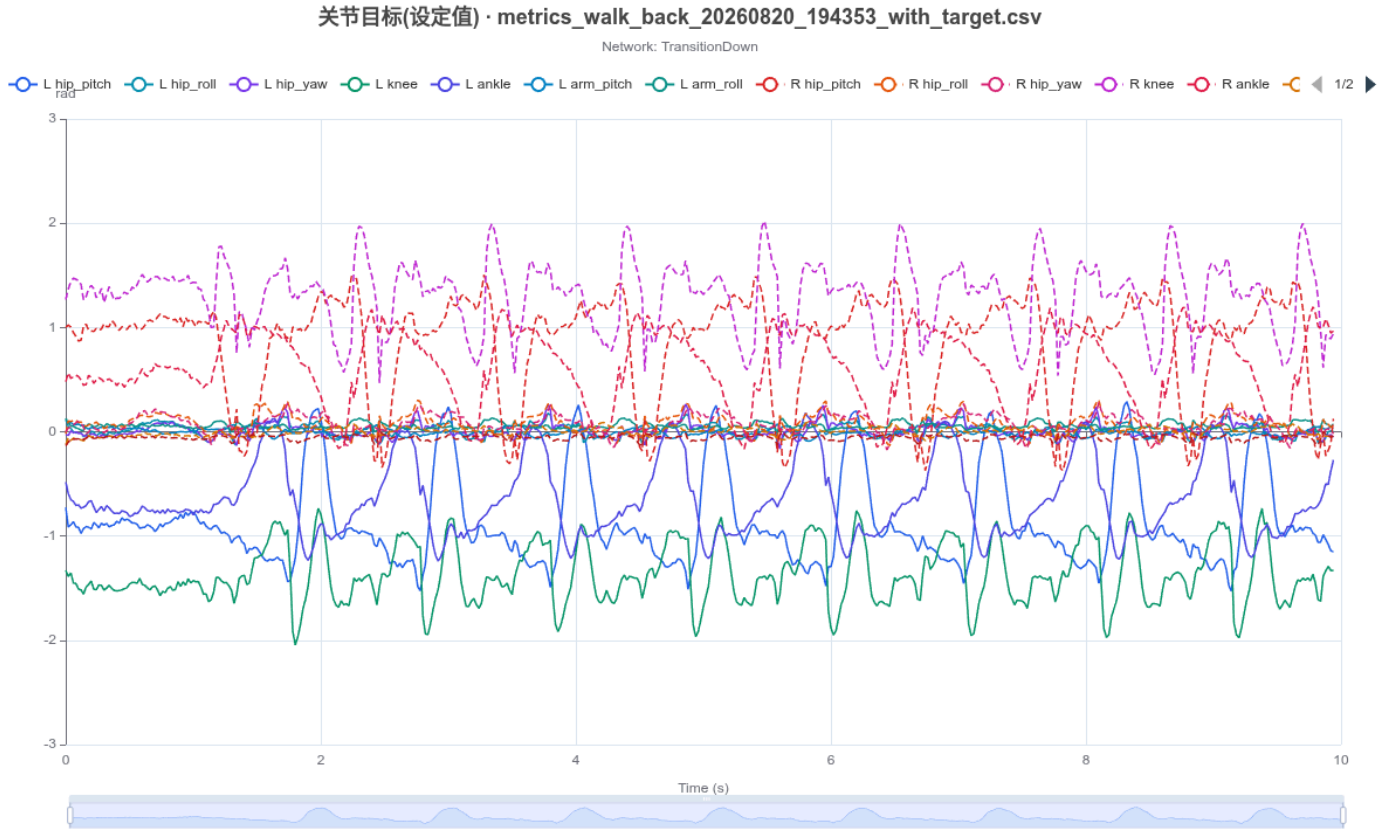


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	498	-1.33943	-0.139593	-0.870726	0.927200	1.33943	1.29256
joint_position_rad/left_hip_roll_joint	10	15	498	-0.0811864	0.0957805	-0.00467731	0.0501673	0.0957805	0.0762920
joint_position_rad/left_hip_yaw_joint	10	15	498	-0.0805884	0.133382	-0.0107396	0.0553086	0.133382	0.0868094
joint_position_rad/left_knee_joint	30	10	498	-1.57014	-1.02578	-1.40399	1.41319	1.57014	1.55436
joint_position_rad/left_ankle_joint	10	15	498	-1.03779	0.0274954	-0.657929	0.730252	1.03779	1.02366
joint_position_rad/left_arm_pitch_joint	10	15	498	-0.0549297	0.0804626	0.000537374	0.0269622	0.0804626	0.0480886
joint_position_rad/left_arm_roll_joint	10	15	498	0.0299933	0.106509	0.0551309	0.0581151	0.106509	0.0957490
joint_position_rad/right_hip_pitch_joint	30	10	498	0.0475623	1.34435	0.860951	0.936085	1.34435	1.29844
joint_position_rad/right_hip_roll_joint	10	15	498	-0.0361721	0.231663	0.0916430	0.118164	0.231663	0.214224
joint_position_rad/right_hip_yaw_joint	10	15	498	-0.0218686	0.237231	0.0943655	0.121016	0.237231	0.214995
joint_position_rad/right_knee_joint	30	10	498	0.920250	1.55975	1.34975	1.36139	1.55975	1.53586
joint_position_rad/right_ankle_joint	10	15	498	-0.0725069	1.04857	0.638097	0.718552	1.04857	1.03213
joint_position_rad/right_arm_pitch_joint	10	15	498	-0.0714377	0.0619537	0.00399352	0.0306026	0.0714377	0.0552144
joint_position_rad/right_arm_roll_joint	10	15	498	-0.0914817	-0.0345112	-0.0550373	0.0563026	0.0914817	0.0783954

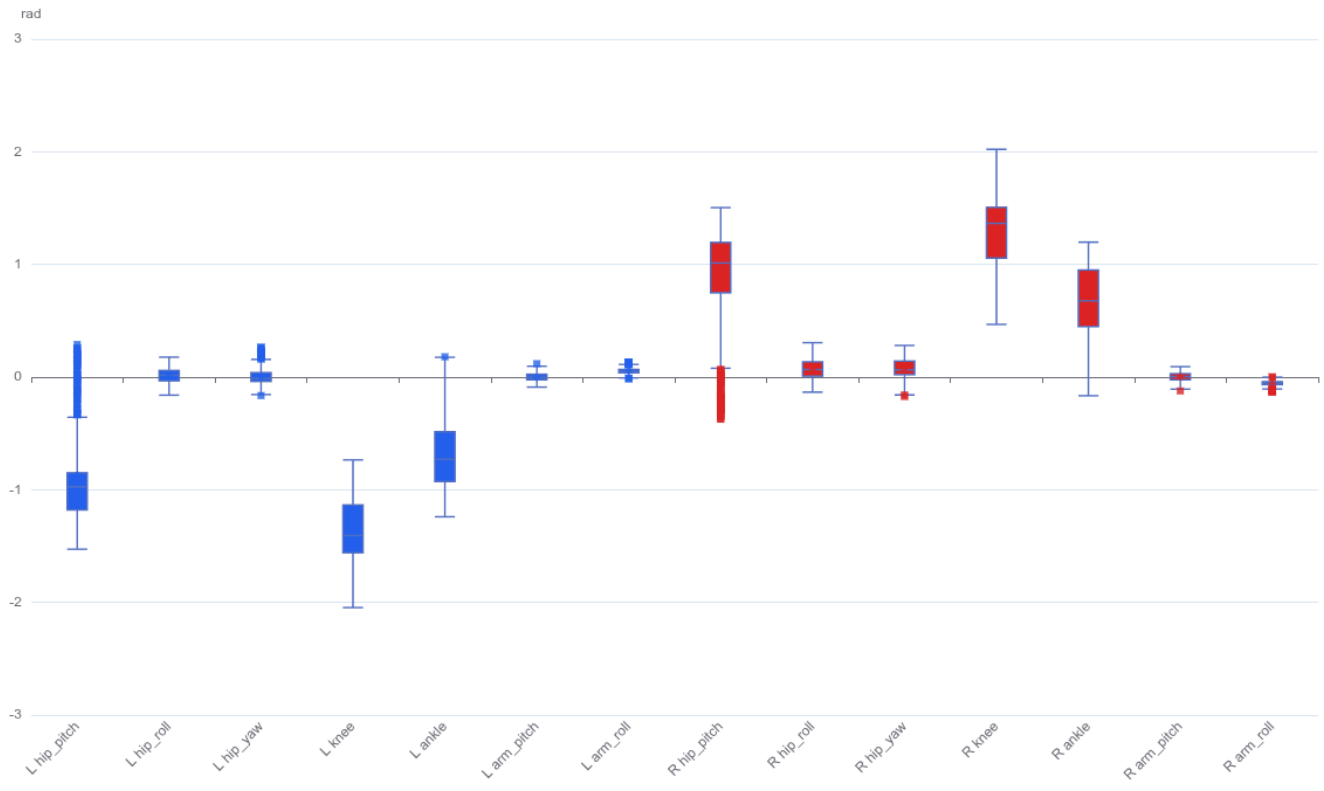

rad
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

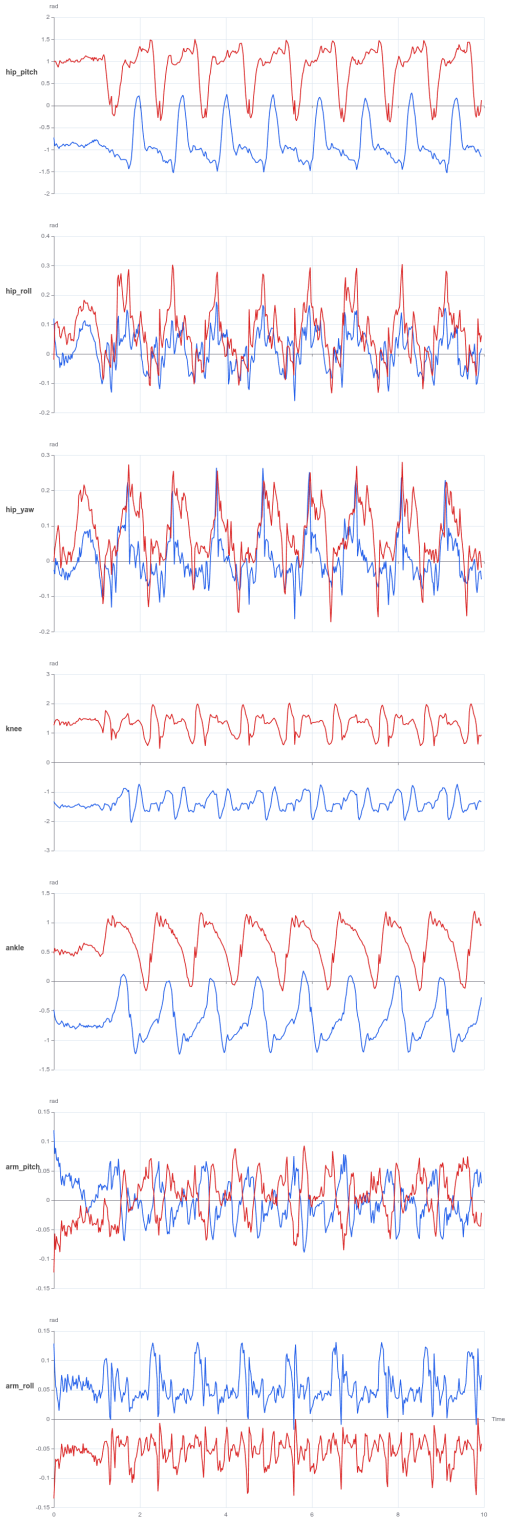

rad
hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节目标(设定值)

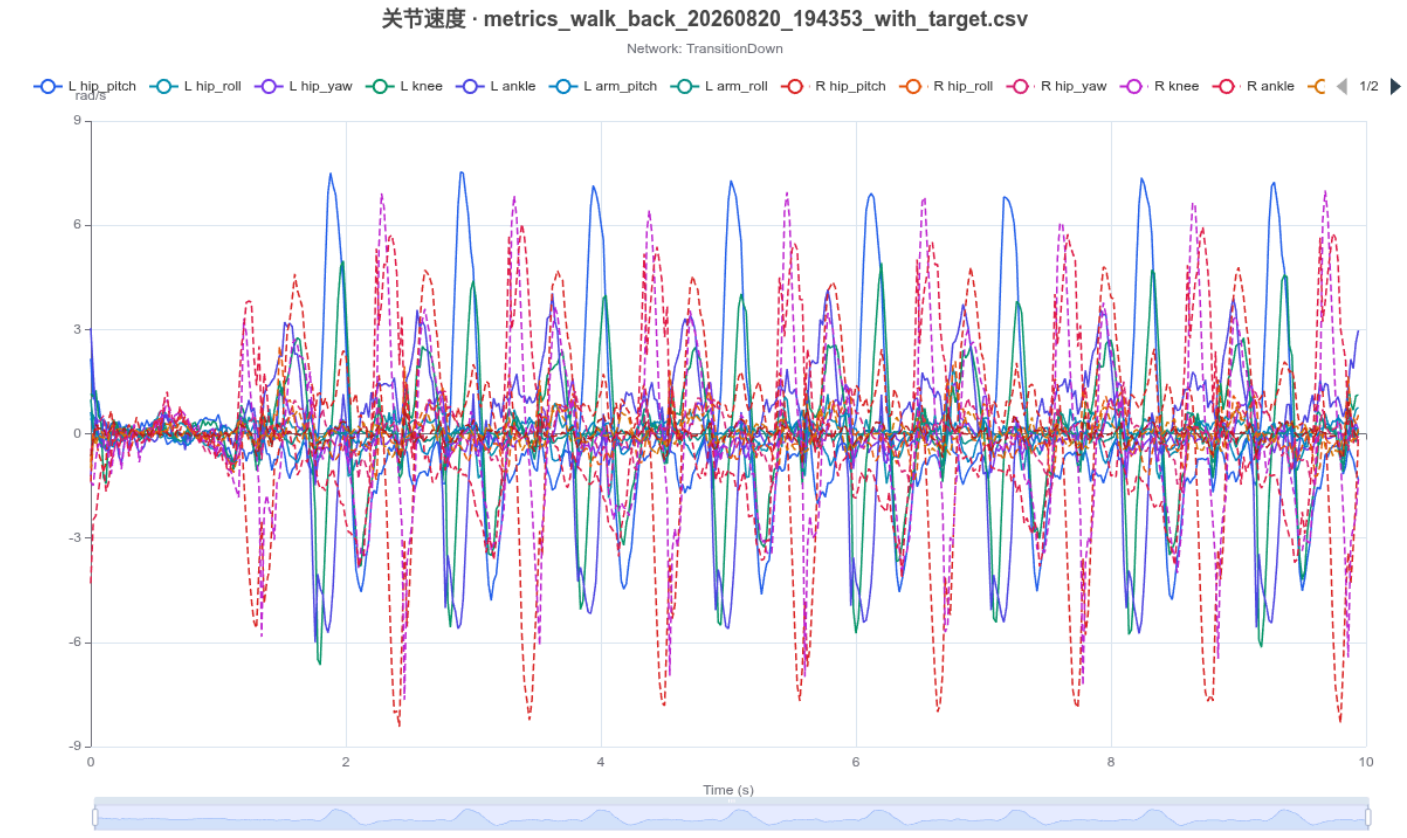


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	498	-1.52782	0.289319	-0.871947	0.972550	1.52782	1.32229
joint_target_rad/left_hip_roll_joint	10	15	498	-0.159942	0.176927	0.0131355	0.0656909	0.176927	0.120239
joint_target_rad/left_hip_yaw_joint	10	15	498	-0.164124	0.264497	0.00454370	0.0681356	0.264497	0.140624
joint_target_rad/left_knee_joint	30	10	498	-2.04548	-0.735249	-1.36712	1.39589	2.04548	1.83058
joint_target_rad/left_ankle_joint	10	15	498	-1.24029	0.180443	-0.654990	0.744076	1.24029	1.13278
joint_target_rad/left_arm_pitch_joint	10	15	498	-0.0884895	0.119347	-0.000406622	0.0341559	0.119347	0.0661799
joint_target_rad/left_arm_roll_joint	10	15	498	-0.0176464	0.131203	0.0554205	0.0621858	0.131203	0.114212
joint_target_rad/right_hip_pitch_joint	30	10	498	-0.374003	1.50488	0.854024	0.987084	1.50488	1.38444
joint_target_rad/right_hip_roll_joint	10	15	498	-0.133548	0.305137	0.0721252	0.114187	0.305137	0.229598
joint_target_rad/right_hip_yaw_joint	10	15	498	-0.172974	0.280905	0.0758587	0.112232	0.280905	0.202936
joint_target_rad/right_knee_joint	30	10	498	0.467876	2.02167	1.29880	1.34112	2.02167	1.85008
joint_target_rad/right_ankle_joint	10	15	498	-0.165216	1.19752	0.641115	0.734502	1.19752	1.08818
joint_target_rad/right_arm_pitch_joint	10	15	498	-0.122484	0.0927826	0.00322002	0.0378141	0.122484	0.0679758
joint_target_rad/right_arm_roll_joint	10	15	498	-0.134602	0.00244593	-0.0550297	0.0593032	0.134602	0.0931620

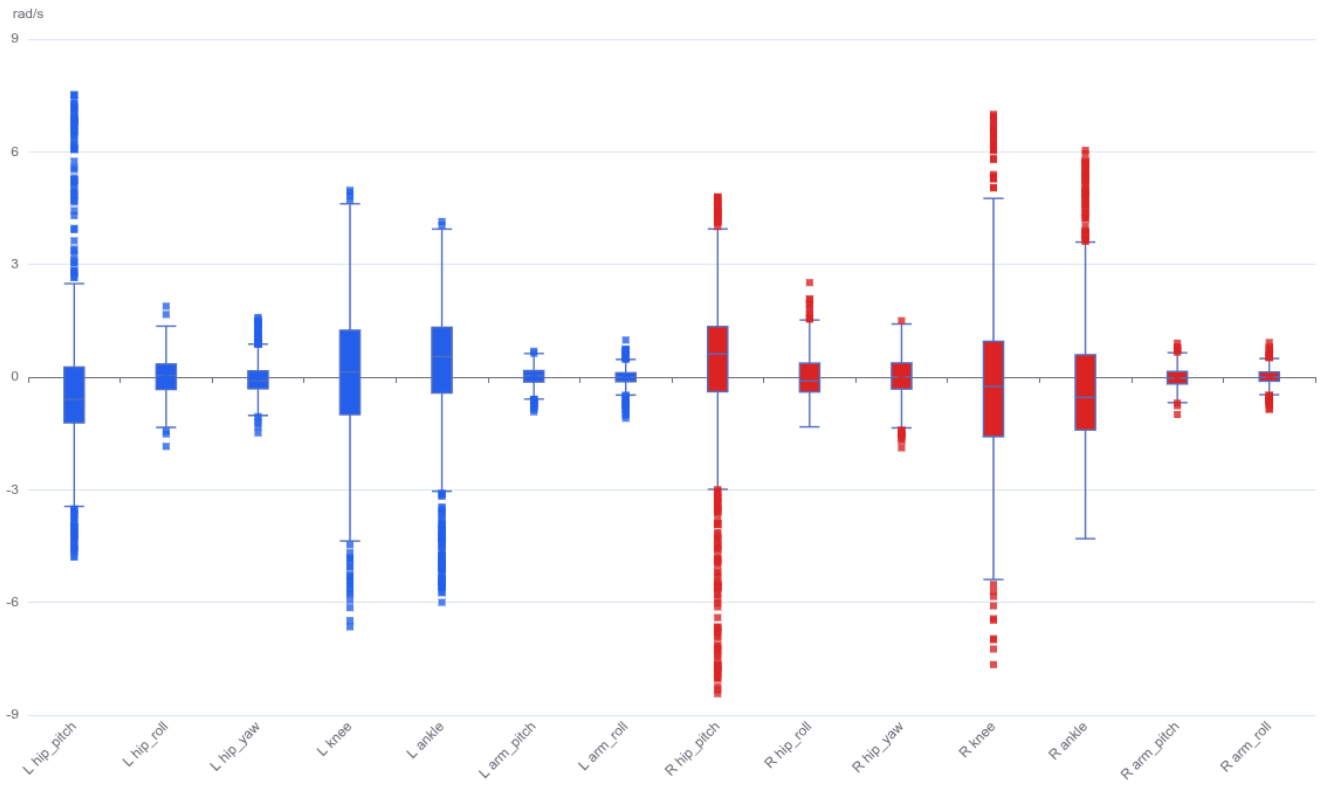

rad
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

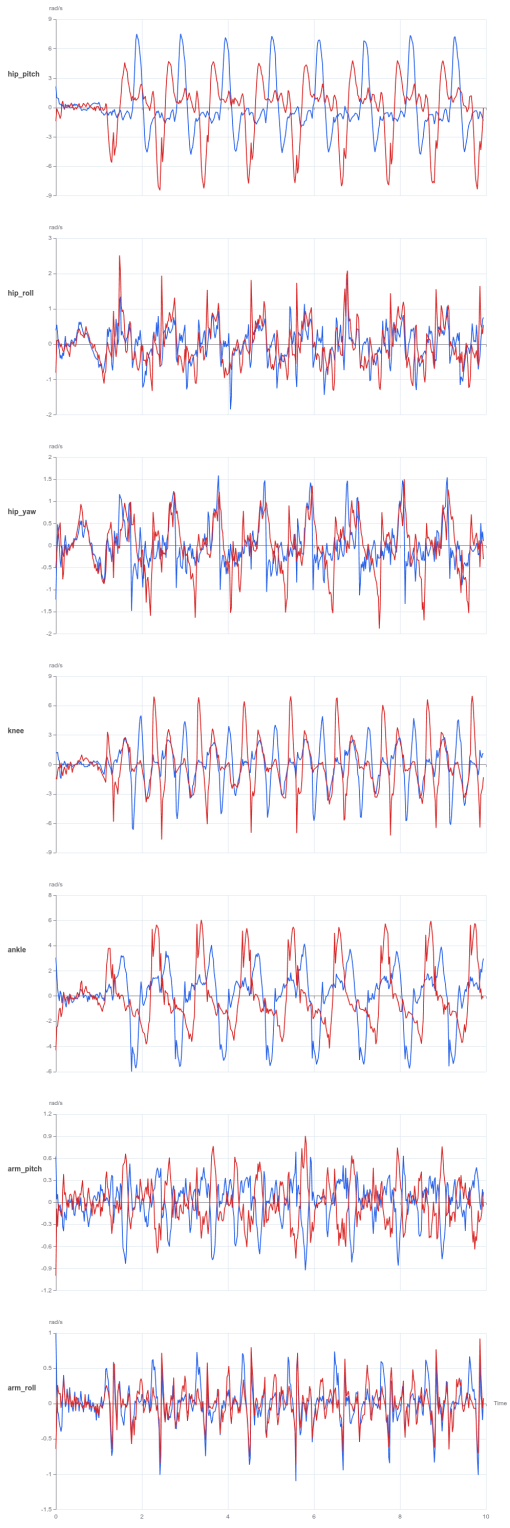

rad
hip_pitch
rad
hip_roll
rad
hip_yaw
rad
knee
rad
ankle
rad
arm_pitch
rad
arm_roll
Time

关节速度

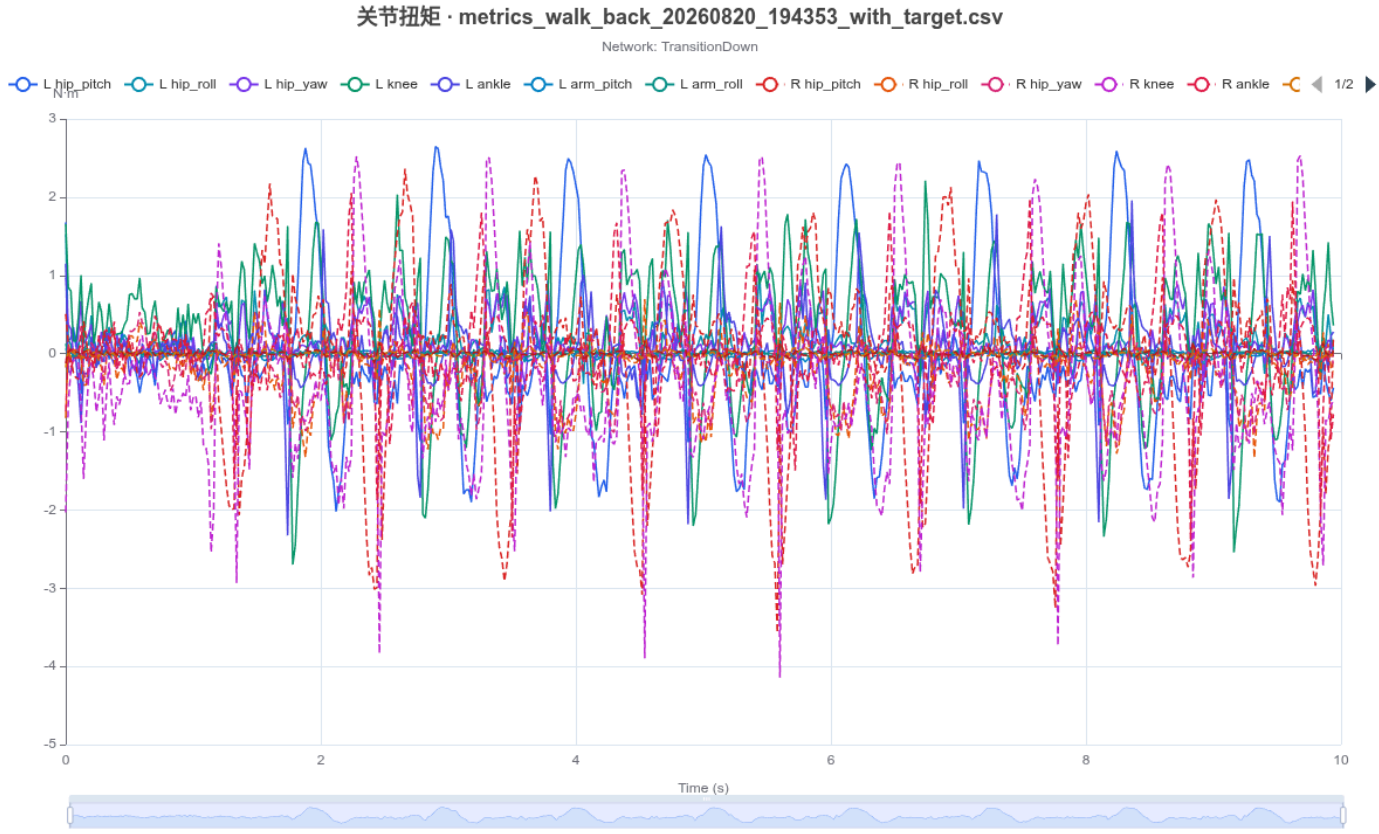


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	498	-4.79377	7.52477	0.00758558	2.68039	7.52477	6.62094
joint_velocity_rad_s/left_hip_roll_joint	10	15	498	-1.84565	1.88737	-0.00453788	0.487803	1.88737	0.980495
joint_velocity_rad_s/left_hip_yaw_joint	10	15	498	-1.48369	1.58862	-0.0121714	0.488766	1.58862	1.05148
joint_velocity_rad_s/left_knee_joint	30	10	498	-6.64977	4.97442	0.00851779	2.10853	6.64977	4.57051
joint_velocity_rad_s/left_ankle_joint	10	15	498	-5.99892	4.14337	0.0282086	2.25882	5.99892	5.01708
joint_velocity_rad_s/left_arm_pitch_joint	10	15	498	-0.922986	0.688917	-0.00235492	0.276266	0.922986	0.617953
joint_velocity_rad_s/left_arm_roll_joint	10	15	498	-1.09573	0.985958	0.00265563	0.278989	1.09573	0.625756
joint_velocity_rad_s/right_hip_pitch_joint	30	10	498	-8.42942	4.80113	-0.0906421	2.92885	8.42942	6.89764
joint_velocity_rad_s/right_hip_roll_joint	10	15	498	-1.32530	2.51504	-0.00347536	0.611596	2.51504	1.11990
joint_velocity_rad_s/right_hip_yaw_joint	10	15	498	-1.88344	1.50130	0.00318894	0.577041	1.88344	1.17800
joint_velocity_rad_s/right_knee_joint	30	10	498	-7.65634	6.99777	-0.0668327	2.49390	7.65634	5.84292
joint_velocity_rad_s/right_ankle_joint	10	15	498	-4.30220	6.03872	0.0121875	2.35172	6.03872	5.15166
joint_velocity_rad_s/right_arm_pitch_joint	10	15	498	-0.996226	0.903428	-0.000804922	0.286965	0.996226	0.579525
joint_velocity_rad_s/right_arm_roll_joint	10	15	498	-0.858983	0.922093	0.00277631	0.240482	0.922093	0.539332

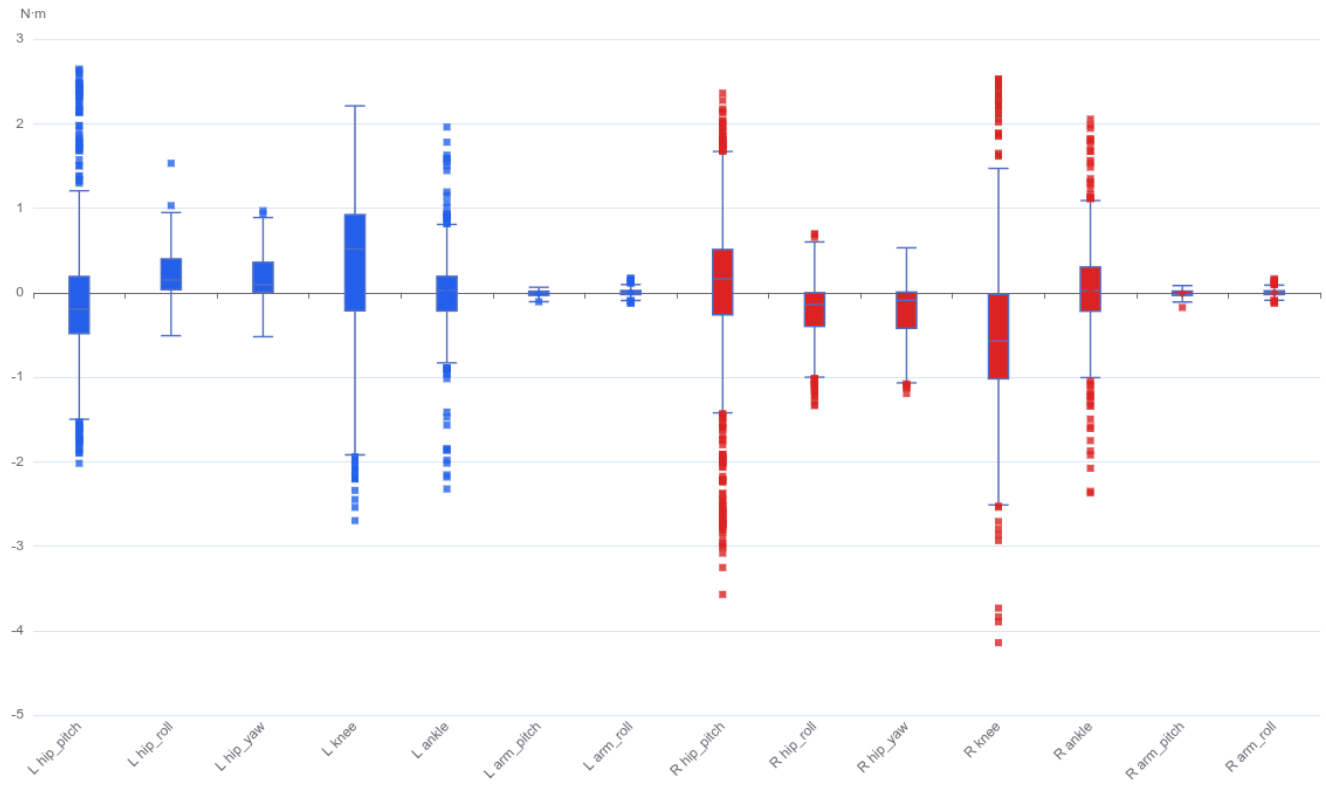

rad/s
9
6
3
0
-3
-6
-9
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

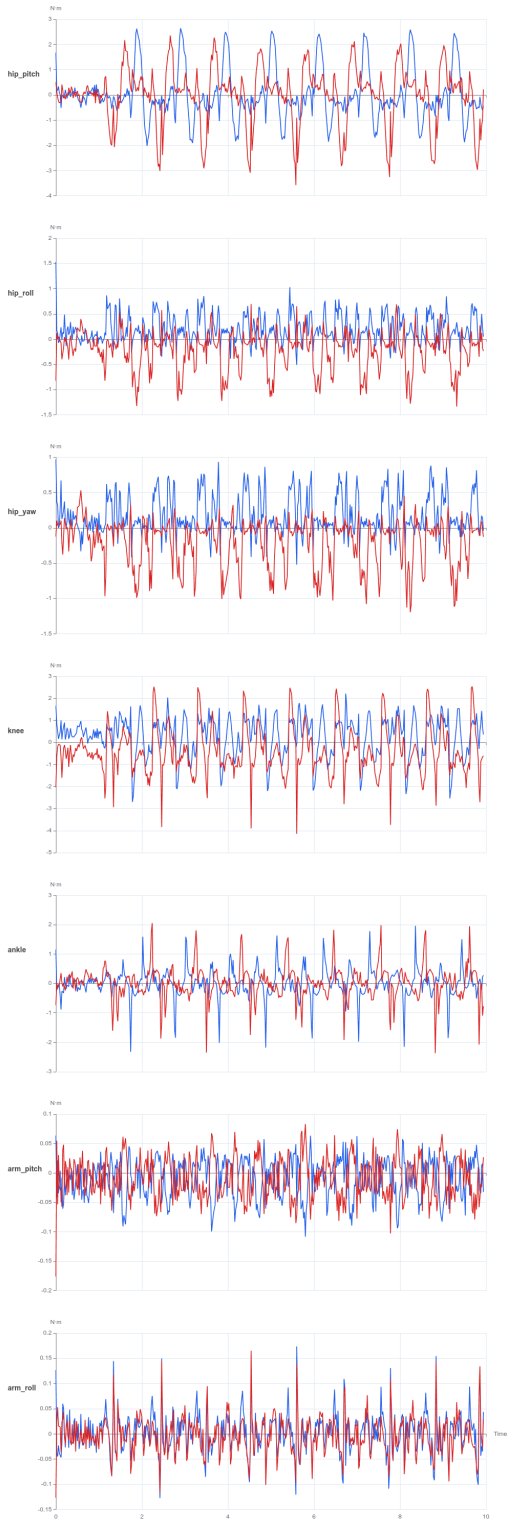

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad/s
Time

关节扭矩

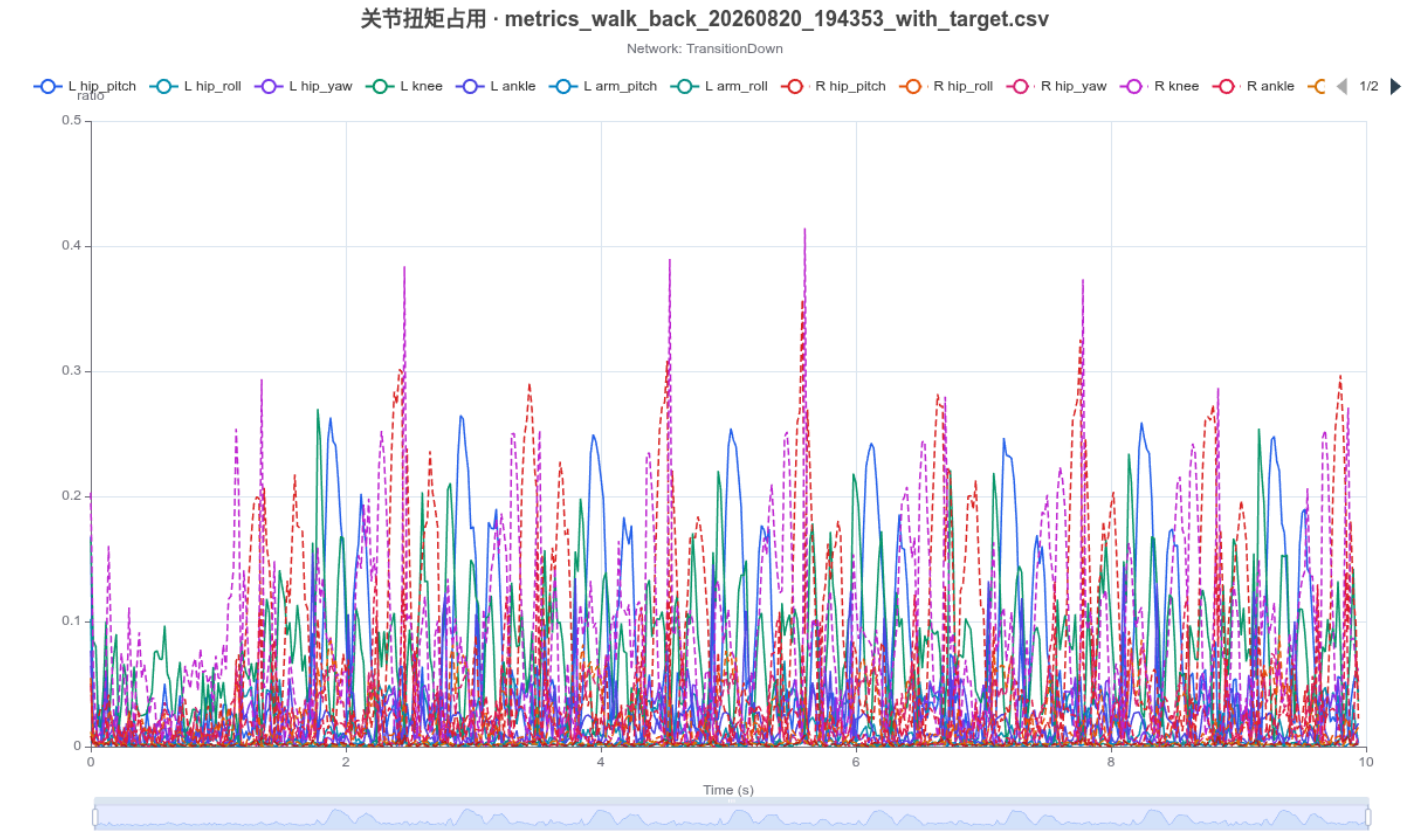


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	498	-2.02023	2.64651	-0.0126522	0.994497	2.64651	2.32821
joint_torque_nm/left_hip_roll_joint	10	15	498	-0.509625	1.53044	0.208098	0.343643	1.53044	0.678372
joint_torque_nm/left_hip_yaw_joint	10	15	498	-0.520970	0.969474	0.182170	0.334709	0.969474	0.722286
joint_torque_nm/left_knee_joint	30	10	498	-2.69838	2.20998	0.314595	0.940036	2.69838	1.70765
joint_torque_nm/left_ankle_joint	10	15	498	-2.32420	1.95929	0.00957415	0.483096	2.32420	0.953058
joint_torque_nm/left_arm_pitch_joint	10	15	498	-0.108174	0.0635611	-0.00896927	0.0345094	0.108174	0.0677712
joint_torque_nm/left_arm_roll_joint	10	15	498	-0.127281	0.173440	0.00152521	0.0401893	0.173440	0.0880914
joint_torque_nm/right_hip_pitch_joint	30	10	498	-3.57180	2.36101	-0.0118326	1.14031	3.57180	2.60714
joint_torque_nm/right_hip_roll_joint	10	15	498	-1.33775	0.696768	-0.224603	0.445239	1.33775	0.990149
joint_torque_nm/right_hip_yaw_joint	10	15	498	-1.19492	0.530790	-0.212843	0.398953	1.19492	0.876961
joint_torque_nm/right_knee_joint	30	10	498	-4.14314	2.52980	-0.416238	1.14118	4.14314	2.23704
joint_torque_nm/right_ankle_joint	10	15	498	-2.37145	2.05317	0.0227552	0.549351	2.37145	1.29059
joint_torque_nm/right_arm_pitch_joint	10	15	498	-0.175972	0.0830470	-0.00890285	0.0364381	0.175972	0.0665683
joint_torque_nm/right_arm_roll_joint	10	15	498	-0.126320	0.164807	-0.00175098	0.0371728	0.164807	0.0766974

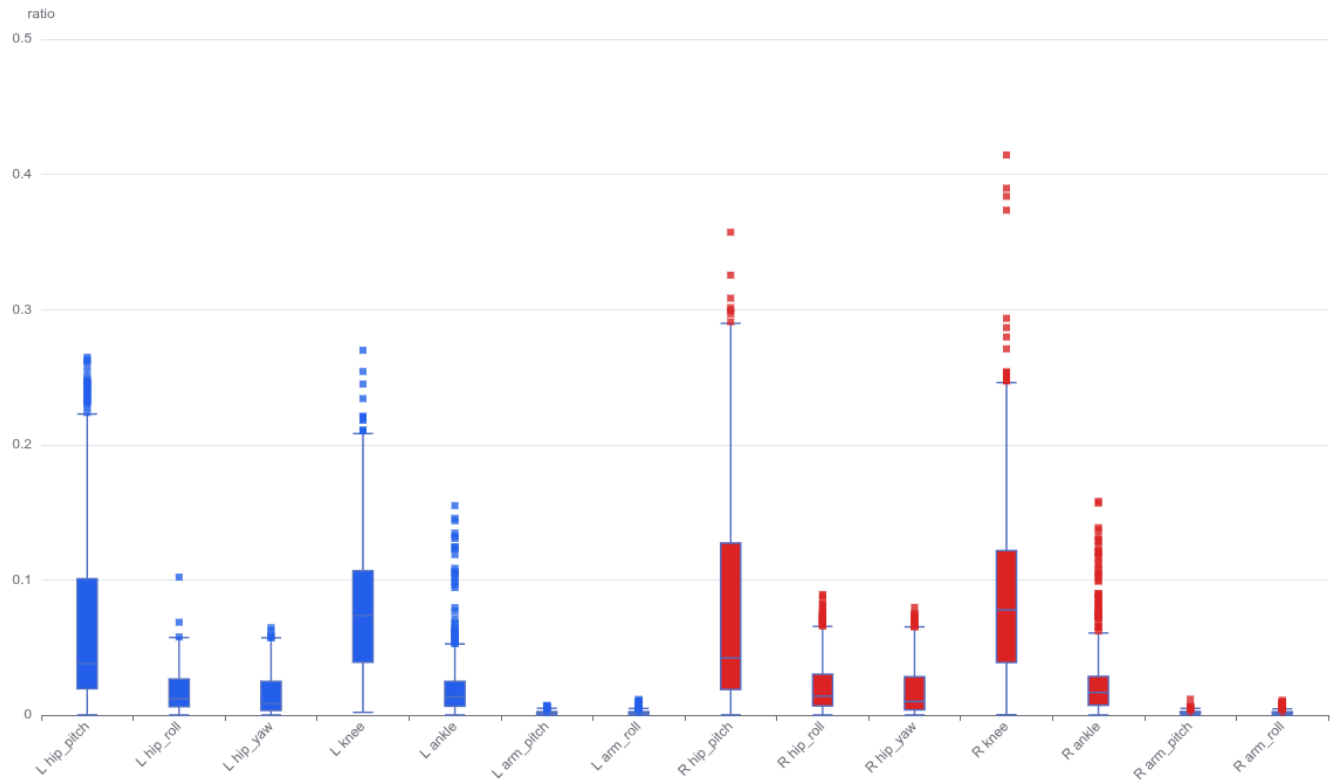

N·m
3
2
1
0
-1
-2
-3
-4
-5
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

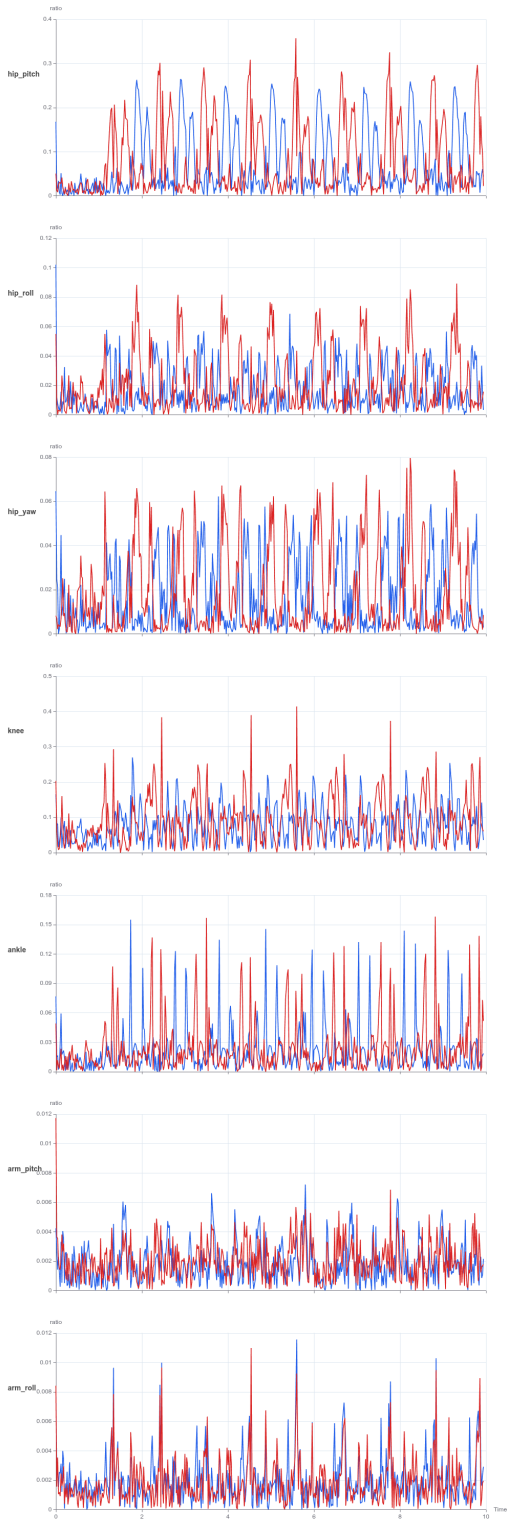

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

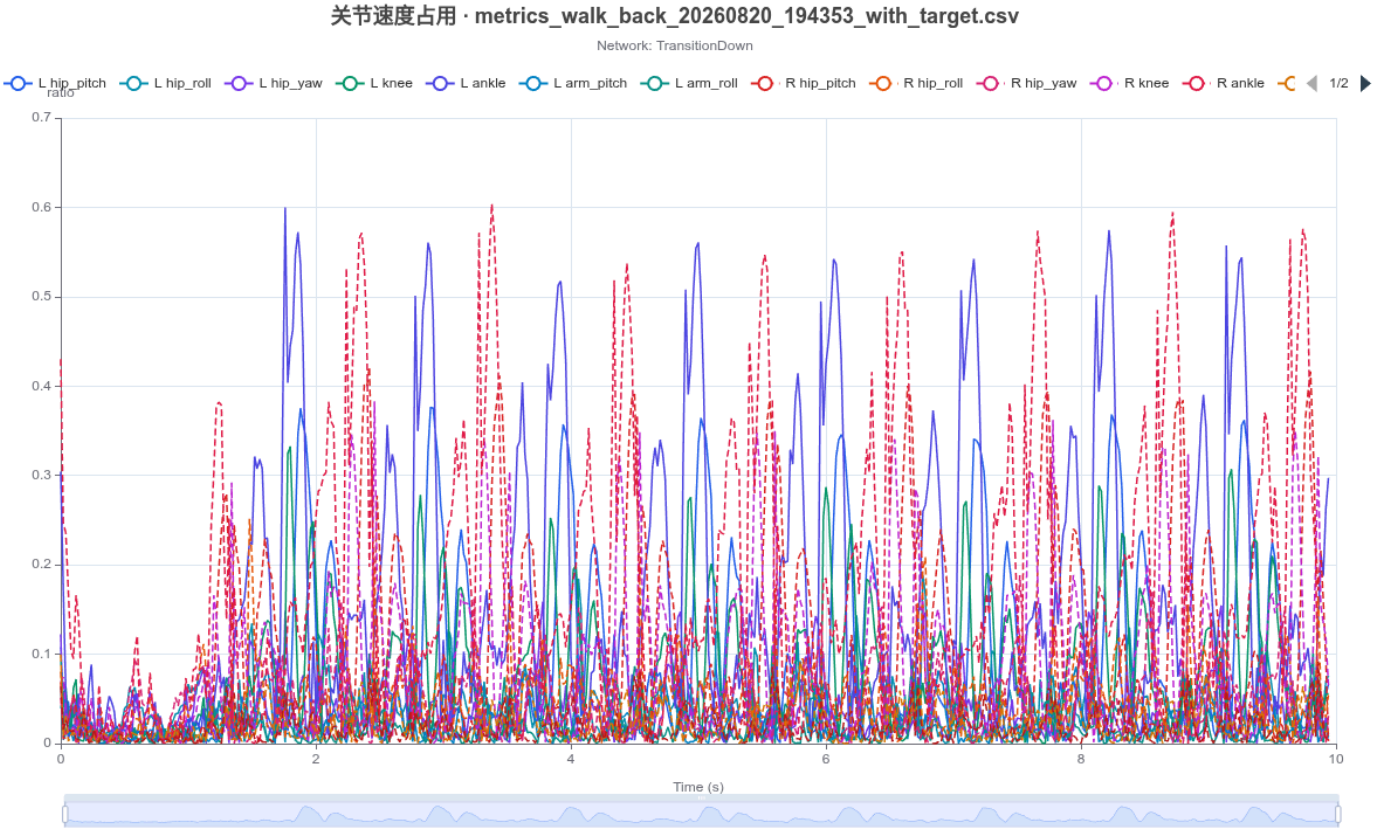


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	498	0.000187742	0.264651	0.0696520	0.0994497	0.264651	0.232821
joint_torque_ratio/left_hip_roll_joint	10	15	498	0.00000857473	0.102029	0.0173906	0.0229096	0.102029	0.0452248
joint_torque_ratio/left_hip_yaw_joint	10	15	498	0.00000988209	0.0646316	0.0158045	0.0223139	0.0646316	0.0481524
joint_torque_ratio/left_knee_joint	30	10	498	0.00201538	0.269838	0.0791260	0.0940036	0.269838	0.170765
joint_torque_ratio/left_ankle_joint	10	15	498	0.0000381121	0.154947	0.0209090	0.0322064	0.154947	0.0635372
joint_torque_ratio/left_arm_pitch_joint	10	15	498	0.00000793069	0.00721163	0.00184665	0.00230063	0.00721163	0.00451808
joint_torque_ratio/left_arm_roll_joint	10	15	498	0.00000908936	0.0115627	0.00202519	0.00267928	0.0115627	0.00587276
joint_torque_ratio/right_hip_pitch_joint	30	10	498	0.000151871	0.357181	0.0802730	0.114031	0.357181	0.260714
joint_torque_ratio/right_hip_roll_joint	10	15	498	0.0000147251	0.0891831	0.0216129	0.0296826	0.0891831	0.0660099
joint_torque_ratio/right_hip_yaw_joint	10	15	498	0.0000188213	0.0796610	0.0184313	0.0265969	0.0796610	0.0584641
joint_torque_ratio/right_knee_joint	30	10	498	0.000287641	0.414314	0.0911320	0.114118	0.414314	0.223704
joint_torque_ratio/right_ankle_joint	10	15	498	0.0000170620	0.158097	0.0244580	0.0366234	0.158097	0.0860392
joint_torque_ratio/right_arm_pitch_joint	10	15	498	0.0000178585	0.0117314	0.00204598	0.00242921	0.0117314	0.00443789
joint_torque_ratio/right_arm_roll_joint	10	15	498	0.0000110724	0.0109871	0.00188938	0.00247819	0.0109871	0.00511316

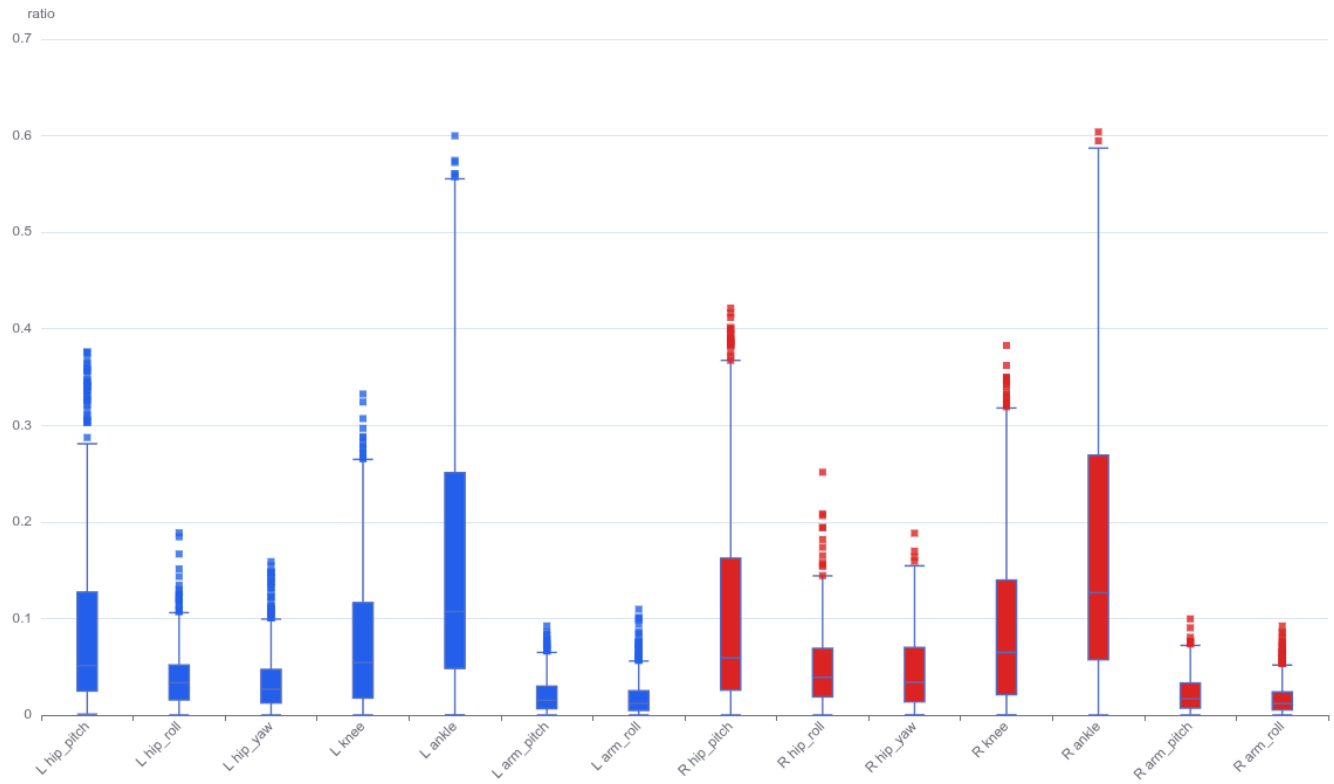

ratio
0.5
0.4
0.3
0.2
0.1
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

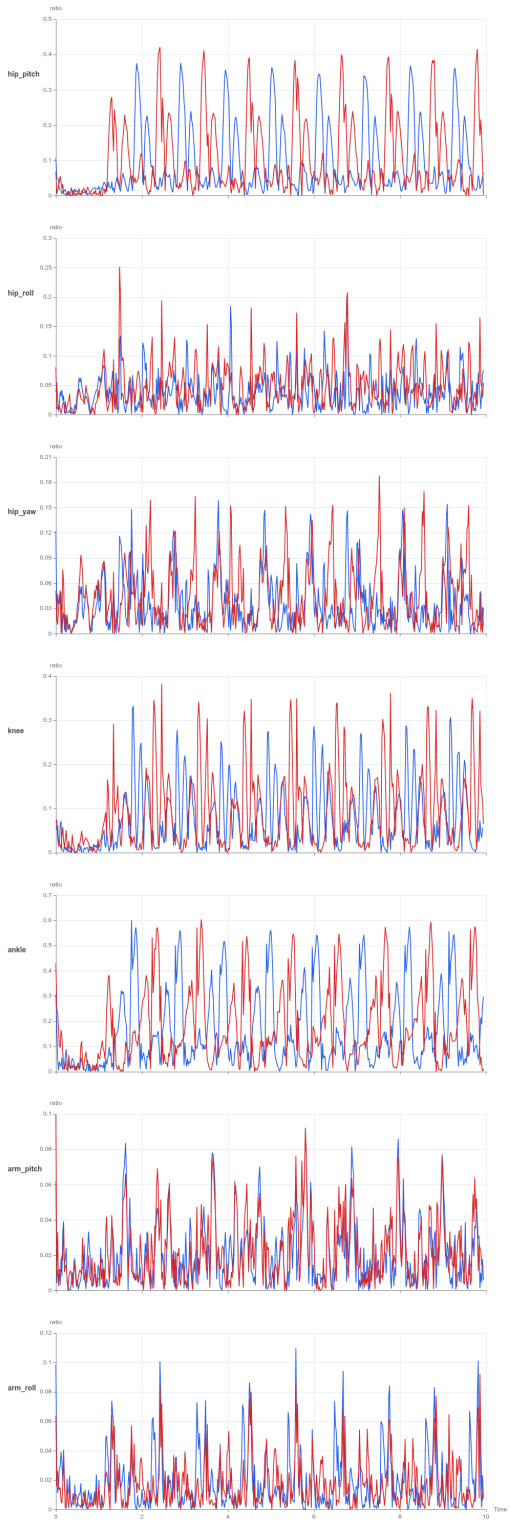

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用

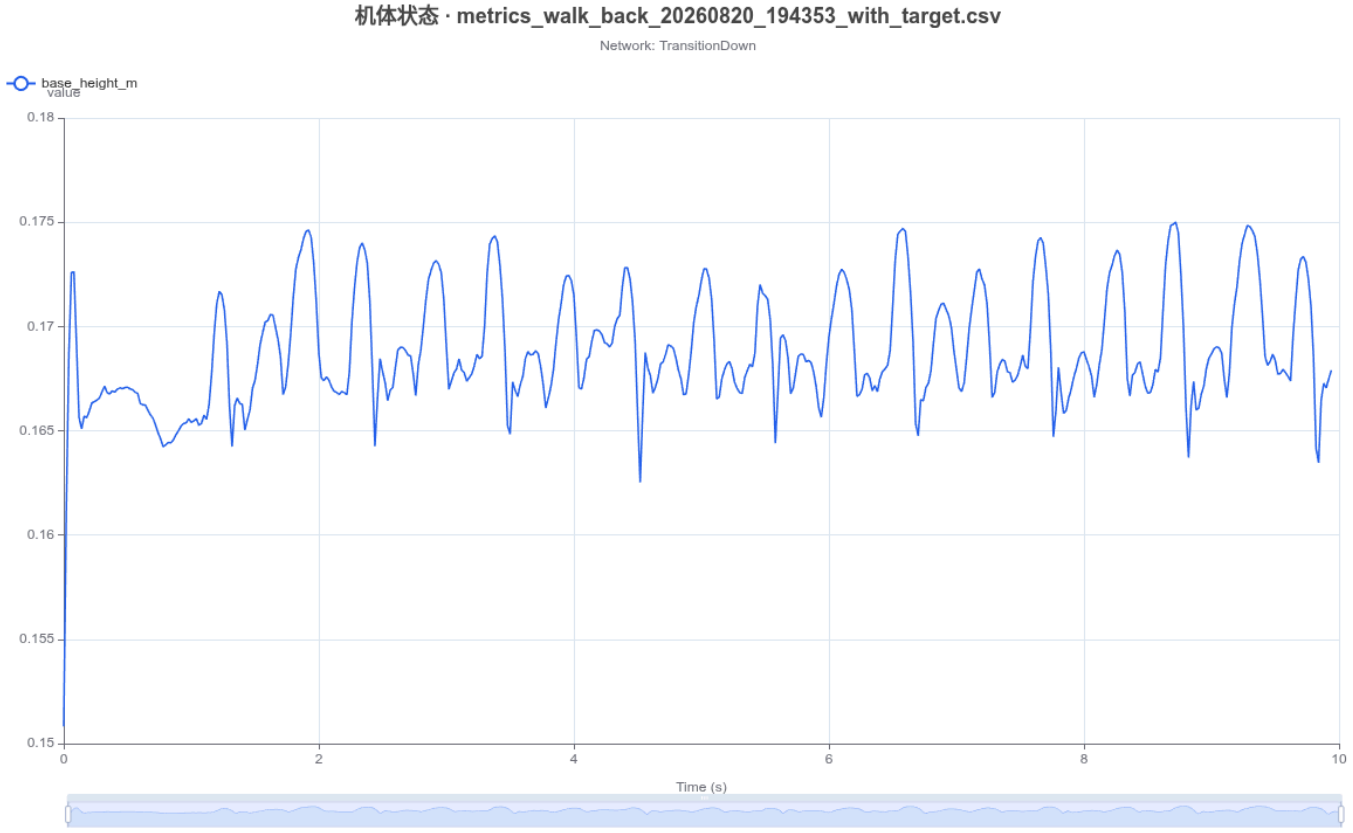


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	498	0.000957080	0.376238	0.0917147	0.134019	0.376238	0.331047
joint_velocity_ratio/left_hip_roll_joint	10	15	498	0.0000160078	0.188737	0.0385128	0.0487803	0.188737	0.0980495
joint_velocity_ratio/left_hip_yaw_joint	10	15	498	0.00000235837	0.158862	0.0362671	0.0488766	0.158862	0.105148
joint_velocity_ratio/left_knee_joint	30	10	498	0.000101946	0.332489	0.0771890	0.105426	0.332489	0.228525
joint_velocity_ratio/left_ankle_joint	10	15	498	0.000179338	0.599892	0.164940	0.225882	0.599892	0.501708
joint_velocity_ratio/left_arm_pitch_joint	10	15	498	0.0000867006	0.0922986	0.0207151	0.0276266	0.0922986	0.0617953
joint_velocity_ratio/left_arm_roll_joint	10	15	498	0.0000190861	0.109573	0.0190744	0.0278989	0.109573	0.0625756
joint_velocity_ratio/right_hip_pitch_joint	30	10	498	0.0000899780	0.421471	0.103172	0.146443	0.421471	0.344882
joint_velocity_ratio/right_hip_roll_joint	10	15	498	0.0000868714	0.251504	0.0478673	0.0611596	0.251504	0.111990
joint_velocity_ratio/right_hip_yaw_joint	10	15	498	0.000195922	0.188344	0.0443833	0.0577041	0.188344	0.117800
joint_velocity_ratio/right_knee_joint	30	10	498	0.000205350	0.382817	0.0906092	0.124695	0.382817	0.292146
joint_velocity_ratio/right_ankle_joint	10	15	498	0.0000758211	0.603872	0.178652	0.235172	0.603872	0.515166
joint_velocity_ratio/right_arm_pitch_joint	10	15	498	0.0000738562	0.0996226	0.0220117	0.0286965	0.0996226	0.0579525
joint_velocity_ratio/right_arm_roll_joint	10	15	498	0.00000828151	0.0922093	0.0173207	0.0240482	0.0922093	0.0539332

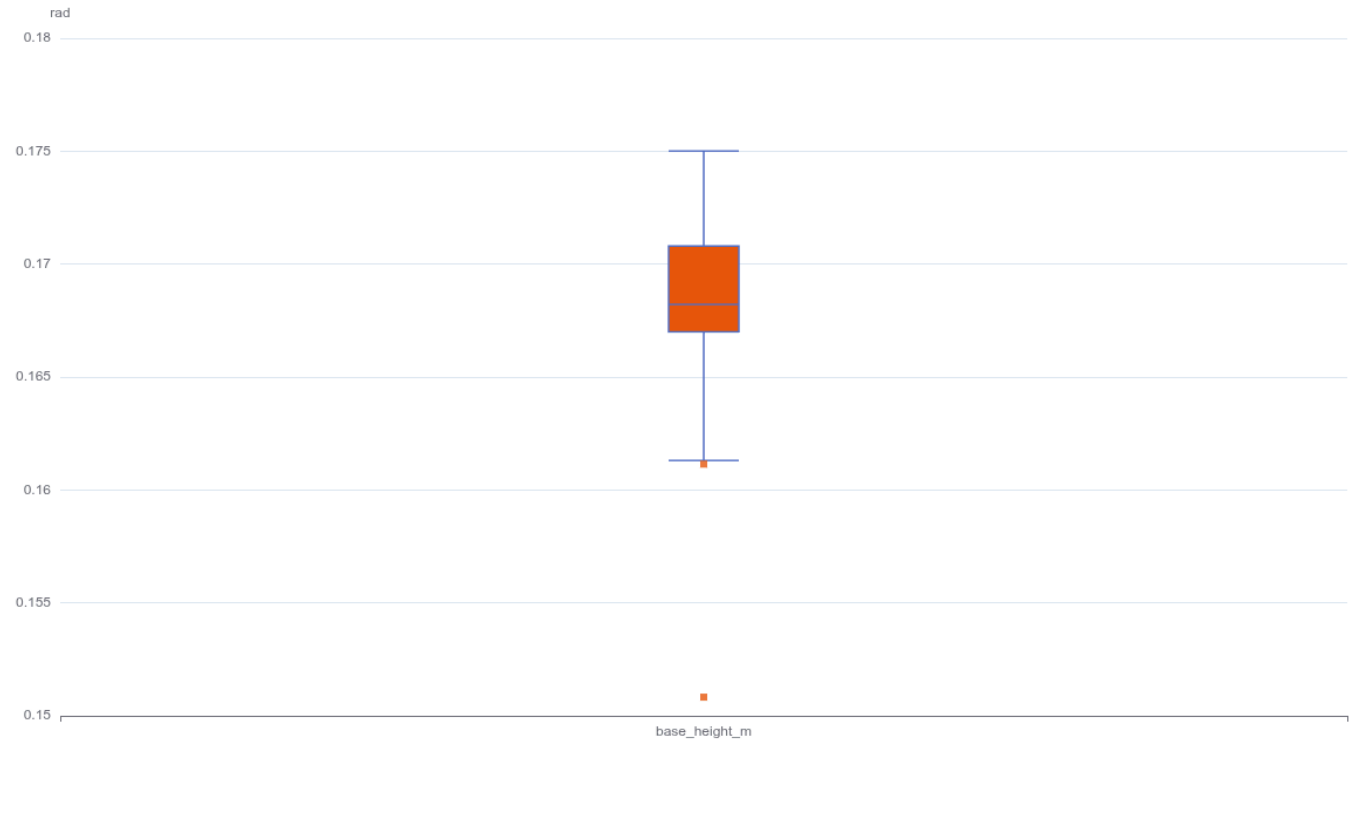

ratio
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

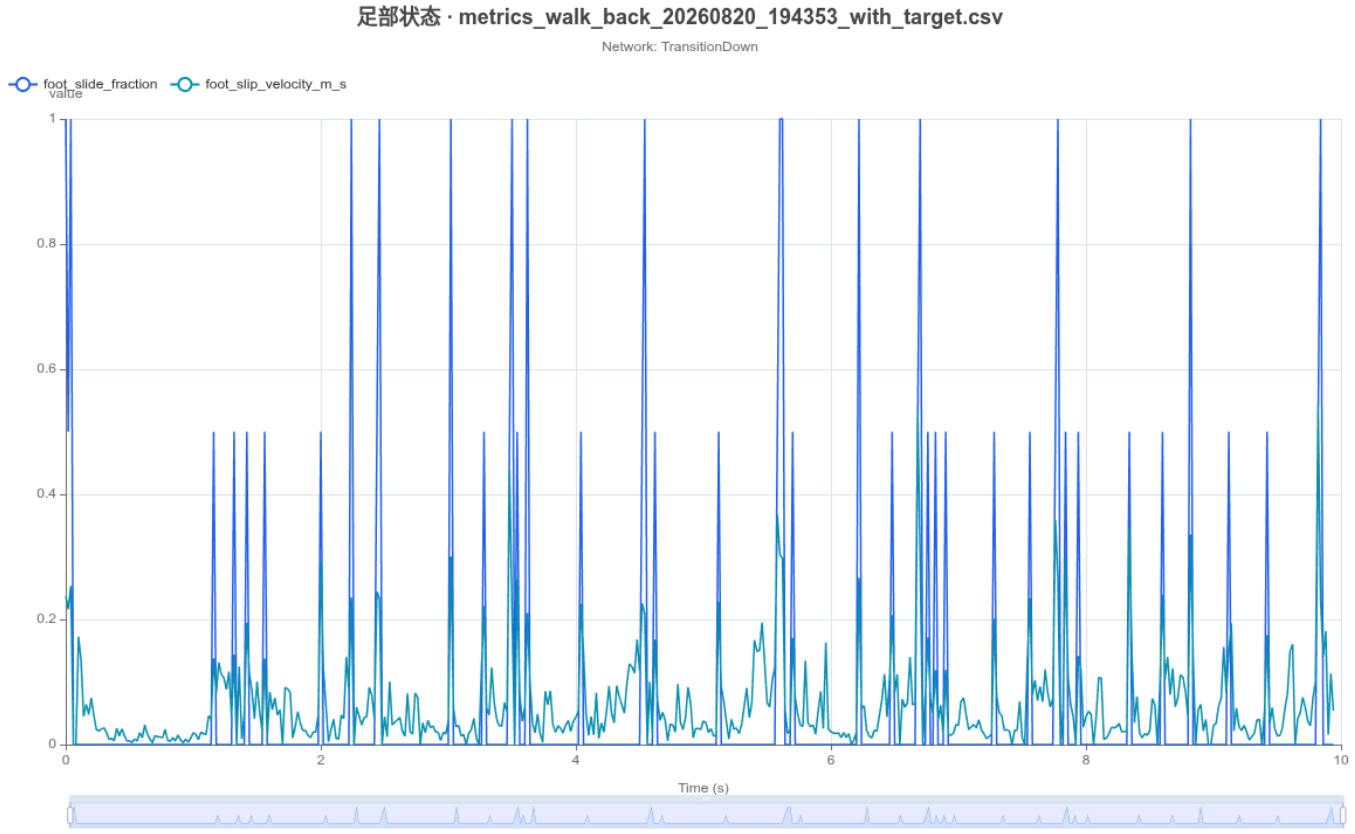
机体状态



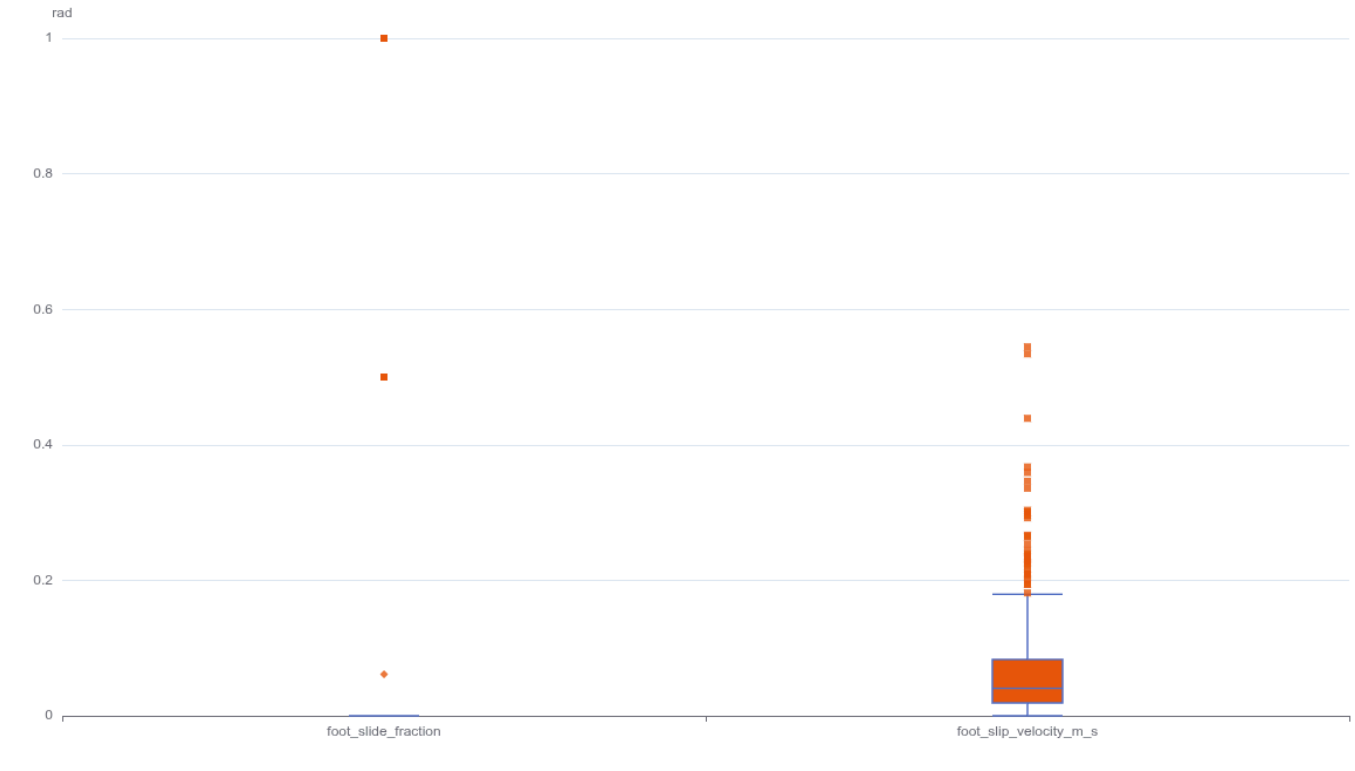
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	498	0.150827	0.175005	0.168842	0.168865	0.175005	0.173994



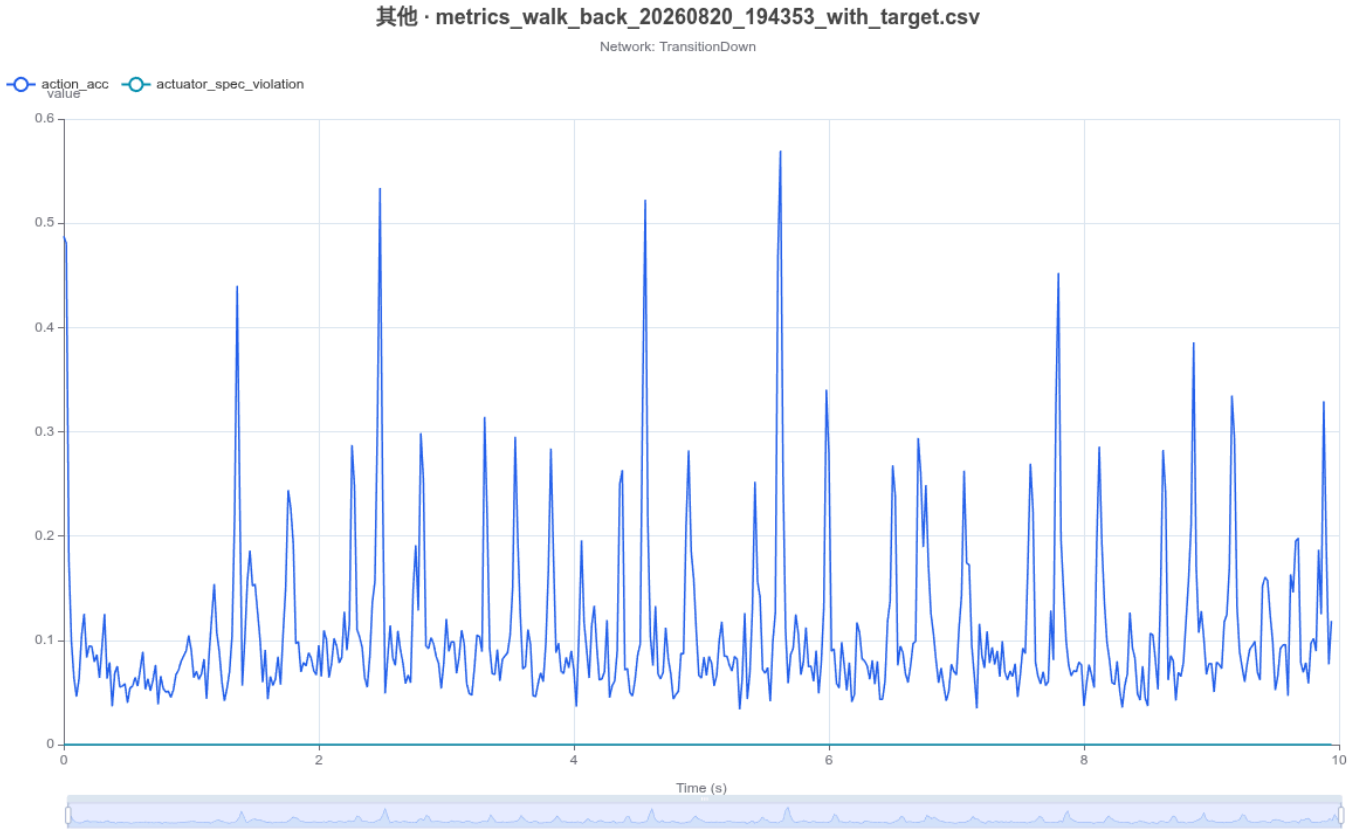
足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	498	0.00000	1.00000	0.0612450	0.213735	1.00000	0.500000
foot_slip_velocity_m_s	–	–	498	0.00000	0.544572	0.0660015	0.0998254	0.544572	0.225724



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	498	0.0338270	0.569515	0.109630	0.135012	0.569515	0.278361
actuator_spec_violation	–	–	498	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

